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EXPERIMENT 10

Aim : To learn Dockerfile instructions, build an image for a sample web application using DOCKERFILE

Theory :

A Dockerfile is a text file that contains a set of instructions used to automate the process of building Docker images. It defines the environment, dependencies, and commands required to run an application inside a container. Docker uses Dockerfiles to ensure consistency, portability, and reproducibility of applications across different systems.

A Dockerfile consists of several important instructions. The FROM instruction specifies the base image (e.g., Ubuntu, Node.js), while WORKDIR sets the working directory inside the container. The COPY or ADD instructions are used to transfer application files into the container. The RUN instruction executes commands during the image build process, such as installing dependencies. The CMD or ENTRYPOINT instruction defines the default command to run the application when the container starts. Additionally, EXPOSE is used to specify the port on which the application will run.

By writing a Dockerfile for a sample web application, developers can build a custom image using the docker build command and run it as a container using docker run. This approach simplifies deployment, ensures environment consistency, and allows the application to run seamlessly across development, testing, and production environments. Docker-based workflows significantly improve scalability and efficiency in modern DevOps practices.

```
C:\Users\Lenovo>docker run redis
1:C 24 Feb 2026 07:57:09.266 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
1:C 24 Feb 2026 07:57:09.266 * Redis version=7.2.4, bits=64, commit=00000000, modified=0, pid=1, just started
1:C 24 Feb 2026 07:57:09.266 # Warning: no config file specified, using the default config. In order to specify a config
 file use redis-server /path/to/redis.conf
1:M 24 Feb 2026 07:57:09.266 * monotonic clock: POSIX clock_gettime
1:M 24 Feb 2026 07:57:09.267 * Running mode=standalone, port=6379.
1:M 24 Feb 2026 07:57:09.268 * Server initialized
1:M 24 Feb 2026 07:57:09.269 * Ready to accept connections tcp
```

```
C:\Users\Lenovo>docker pull redis
Using default tag: latest
latest: Pulling from library/redis
0c8d55a45c0d: Pull complete
9a00fba7918f: Pull complete
253eb0a88de9: Pull complete
2ccab22a2cd8: Pull complete
5a312569f2e0: Pull complete
4f4fb700ef54: Pull complete
fb0cbf9357d9: Pull complete
Digest: sha256:dc123d5a424cbe44d681822c2a620f3f44bdba240b8a58a0eee0048a285532ed
Status: Downloaded newer image for redis:latest
docker.io/library/redis:latest

What's Next?
  1. Sign in to your Docker account → docker login
  2. View a summary of image vulnerabilities and recommendations → docker scout quickview redis
```

```
C:\Users\Lenovo>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

```
C:\Users\Lenovo>docker ps
CONTAINER ID   IMAGE                                    COMMAND                  CREATED         STATUS          PORTS
      NAMES
1c17503a759c   170a1e90f843                             "docker-entrypoint.s…"   5 minutes ago   Up 22 seconds   6379/
tcp   amazing_robinson
a388ed3e2ae7   docker/dev-environments-default:stable-1 "sleep infinity"         23 months ago   Up 26 seconds
      siddhesh-app-1
```

```
C:\Users\Lenovo>docker stop 1c17503a759c
1c17503a759c

C:\Users\Lenovo>docker start 1c17503a759c
1c17503a759c
```

```
C:\Users\Lenovo>docker rm 1c17503a759c
1c17503a759c
```

```
C:\Users\Lenovo>docker images
REPOSITORY                        TAG                                       IMAGE ID       CREATED         SIZE
redis                             latest                                    f0acdeb5ced3   14 hours ago    140MB
ubuntu                            latest                                    802541663949   6 months ago    78.1MB
ubuntu                            <none>                                    ca2b0f26964c   24 months ago   77.9MB
redis                             <none>                                    170a1e90f843   2 years ago     138MB
docker/dev-environments-default   stable-1                                  7eb85f44c229   3 years ago     610MB
crazymax/linguist                 7.20.0                                    dc5f9ae3be6a   3 years ago     72.2MB
docker/desktop-git-helper         5a4fca126aadcd3f6cc3a011aa991de982ae7000  efe2d67c403b   4 years ago     44.2MB
```

```
C:\Users\Lenovo>docker rmi f0acdeb5ced3
Untagged: redis:latest
Untagged: redis@sha256:dc123d5a424cbe44d681822c2a620f3f44bdba240b8a58a0eee0048a285532ed
Deleted: sha256:f0acdeb5ced3472faa8a41497498730d1b30e52ab4b8bf867b5b5ddb87720ec6
Deleted: sha256:ef393d323f971e457b565382266ff95d29e75cdcb8fa437e9bf7b73fc7313c11
Deleted: sha256:7ca2926f5c45fc35d3d02d66ad386ff0e4067527df8ca12b1b2189b185ddcfbc
Deleted: sha256:365e4cce10e83a6bb3cf41f881e4869ab7cdbaa65371796f090558e1ca2aae7e
Deleted: sha256:42a797d73e006fc25a5ca6dc274c835c061f206832948154875f994fabfb2d50
Deleted: sha256:fba38928ef78663422aca98187e21a3f0012559ec1a73c776b2de370967c45d3
Deleted: sha256:b599de35d6f51c157a9d80d634e4f2f4751301df3d2498f5acb35167a8d34642
Deleted: sha256:a8ff6f8cbdfd6741c10dd183560df7212db666db046768b0f05bbc3904515f03
```

```
C:\Users\Lenovo>docker run --name ubuntu --rm -i -t ubuntu bash
root@0561ed697d99:/# exit
exit
```

```
C:\Users\Lenovo>docker run -d -p 8080:8080 ubuntu
f2de7b7b95eddddf5d839545d2e360376f90dee740084e1874c5c2c103bdcf1c
docker: Error response from daemon: Ports are not available: exposing port TCP 0.0.0.0:8080 -> 0.0.0.0:0: listen tcp 0.0
.0.0:8080: bind: Only one usage of each socket address (protocol/network address/port) is normally permitted.
```

Coclusion : Successfully implemented Dockerfile instructions, build an image for a sample web application using DOCKERFILE